Inequalities — problem 7.58

A. Engel — Problem-Solving Strategies · 1998

Let x, y, z be the lengths of the sides of a triangle, and let

$$f(x,y,z) = \left| \frac{x-y}{x+y} + \frac{y-z}{y+z} + \frac{z-x}{z+x} \right|.$$

Prove that (a) $f(x,y,z) < 1$. (b) $f(x,y,z) < 1/8$. (c) Find the upper limit of $f(x,y,z)$.